

Ethan J. Guerrero

San Antonio, TX — (210) 480-8675 — ethanguerrero2814@gmail.com — Portfolio

Education

Texas A&M University

Bachelor of Science in Biomedical Engineering

Cumulative GPA: 3.24/4.00

College Station, TX

May 2026

Certifications & Training

- **Good Clinical Practice (GLP, Device GCP, FDA-Focused GCP, Social/Behavioral GCP)** — CITI Program — 2025–2028
- **Responsible Conduct of Research & IACUC Training** — CITI Program — 2025–2028
- **Registered Pharmacy Technician** — Texas State Board of Pharmacy — 2025–2028
- **Certified Pharmacy Technician (CPhT)** — National Healthcareer Association — 2025–2028
- **SolidWorks: Sketching & Finite Element Analysis** — LinkedIn Learning — 2025

Awards & Honors

- **Bullet Pitch Competition, 1st Place (2026)** — Won \$100 by delivering a one-minute, no-visual pitch for a canine vital-signs monitoring wearable vest.
- **MedXPlore Competition, 2nd Place (2026)** — Developed and pitched VertaVolt, a piezoelectric spinal-stimulator concept designed to support treatment of chronic neuropathic pain, following 48 hours of medical-device market, problem, and regulatory research.
- **Kimberly Mae Wilcox Memorial Scholarship (2025)** — Awarded by the Texas A&M Biomedical Engineering Department in recognition of academic dedication, leadership, and departmental involvement.
- **Aggie Pitch Finalist (2024)** — Selected as 1 of 10 finalists after interviewing 14 pharmacy technicians and 8 pharmacists, then pitching an affordable gravity-fed pill-filling concept to reduce prescription-processing bottlenecks in retail pharmacies.
- **Linda & Joe Fowler '68 Scholarship (2024)** — Awarded by the Texas A&M Biomedical Engineering Department in recognition of leadership, extracurricular involvement, and academic dedication.

Skills

- **Programming Languages:** Python, SQL, MATLAB, HTML, JavaScript
- **Libraries & Frameworks:** React, React Native, Pandas, scikit-learn, Matplotlib, Selenium
- **Developer Tools:** Git, GitHub, SQLite, Linux
- **Engineering & Lab Tools:** SolidWorks, AutoCAD, LabVIEW, Multisim, Minitab, ImageJ, PyMOL
- **Productivity Tools:** Microsoft Office Suite, Google Workspace

Academic Projects

MedBuddy — Medication Adherence Mobile App

Mobile App Developer, Senior Capstone Team

Aug 2025 – May 2026

College Station, TX

- Developed the mobile app for MedBuddy, pairing a smart pill dispenser with gamified reminders to target medication non-adherence, a problem tied to an estimated \$300B in annual U.S. healthcare costs; authored design inputs and acceptance criteria mapped to usability (ISO 9241-210), use-error (IEC 62366), and software-lifecycle (IEC 62304) standards.
- Interviewed 17 users and reviewed 10 medication-adherence studies to define requirements for an RFID-based medication-monitoring system paired with a mobile app and provider dashboard.

- Coded and executed the app-side verification protocol (70 simulated dose events, 100 concurrent leaderboard-load events), ran a 14-day, 10-user field study, and analyzed usability-survey (n=11) and interaction-log data to drive improvements.

Hospital Readmission Prediction Model

March 2026

Individual Project, BMEN 351

College Station, TX

- Cleaned and engineered a 101,766-encounter diabetic-patient dataset from 130 U.S. hospitals down to 89 predictive features, handling missing values and encoding categorical variables for modeling.
- Built and compared four ML models in Python — KNN, Random Forest, Multinomial Logistic Regression, and a sequential neural network — to predict 30-day hospital readmission risk in diabetic patients.
- Achieved 74.70% test accuracy with the top model, a 20.8-point lift over the 53.9% majority-class baseline, validated with ROC curves and class-level metrics.

BMEN 344: Biological Interactions & Testing Laboratory

Fall 2025

Texas A&M University

College Station, TX

- Evaluated the in vitro biocompatibility of copper, HDPE, alumina, SS316L stainless steel, and ABS by exposing L929 murine fibroblasts to each material and assessing cytotoxicity through complementary qualitative and quantitative methods.
- Cultured L929 murine fibroblasts in 24-well plates and cryopreserved cell stocks to maintain consistent cell lines across testing replicates.
- Measured biomaterial-extract absorbance via spectrophotometry and analyzed fluorescence-microscopy images in ImageJ to assess cell morphology and viability following material exposure.
- Identified copper as severely cytotoxic (grade 3) and SS316L as mildly cytotoxic (grade 1), while HDPE, alumina, and ABS were non-cytotoxic (grade 0), by applying ISO 10993-5 direct-contact testing to L929 fibroblasts for 7 days across 3 replicates per material.

BMEN 343: Biomedical Engineering Materials Laboratory

Spring 2025

Texas A&M University

College Station, TX

- Investigated copper surface modification of SS316L stainless steel to reduce bacterial adhesion and biofilm formation on mechanical heart-valve surfaces, addressing the risk of prosthetic-valve endocarditis and device-associated infection.
- Electroplated, air-dried, and annealed copper coatings on SS316L; evaluated coating uniformity by optical microscopy, wettability by contact-angle testing, and protein adsorption by Bradford assay, identifying annealed copper as the most hydrophobic surface at 125.09° versus 73.28° for uncoated stainless steel.
- Verified plasma-induced surface functional groups (hydroxyl, carbonyl, carboxyl) on UHMWPE via FTIR-ATR analysis, linking increased surface hydrophilicity to reduced contact angles and higher Bradford-assay protein adsorption.
- Investigated degradation of SS316L, alumina, and UHMWPE tibial-fracture bone-plate materials under simulated physiological (acidic) conditions per ISO 10993-18, assessing mass loss, compression strength, and surface morphology; found UHMWPE most susceptible to mechanical degradation while SS316L and alumina retained structural integrity.

Biomedical Mechanics Laboratory

Aug 2024 – Dec 2024

Texas A&M University

College Station, TX

- Evaluated finite element simulations comparing stress, force distribution in intact and plated bone models.
- Assisted in compression testing to validate simulation responses against experimental measurements for the plated-bone-model analysis.
- Examined vascular stent mechanical performance by analyzing deformation and stress under load.

Personal Projects

Web Scraper & Recommendation Engine

May 2026 – Present

Individual, Personal Project

College Station, TX

- Built a Selenium-based Python web scraper that collected and stored 2,700+ targeted job postings in a relational SQL database, removing 674 duplicate postings and optimizing queries for faster filtering.
- Applied rule-based and ML/BM25 scoring — including word tokenization — to classify and rank listings by role type and seniority level, then layered in prompt engineering and retrieval-augmented generation (RAG) so users can search postings matching their interests in natural language.
- Designed the system architecture and data flow across scraping, validation, storage, and retrieval components, documenting methods and decisions to support future variations.

Personal Finance Analytics Tool

May 2026 – Present

Individual, Personal Project

College Station, TX

- Built a Python tool aggregating 3,000+ financial data points to track income, expenses, and investments.
- Researched project requirements, selected data-management, analytical tools for financial analysis, then validated transaction amounts, dates, account balances against official financial statements for accuracy.
- Migrated data storage from flat CSV files to a normalized SQLite database, improving query performance and data integrity for ongoing analysis.
- Built a React dashboard connected through a Python API backend to graphically visualize income, expenses, and account balances, with live updates on refresh instead of re-running the underlying script.

Canine Vitals Monitoring Vest, Project Lead

Jan 2025 – May 2026

TURTLE Robotics, Texas A&M University

College Station, TX

- Founded and led an 18-member team developing a wearable vest to monitor canine vital signs, reviewing 80+ applications and interviewing 12 candidates to build the team.
- Researched and selected sensor options for accurate, comfortable vital-sign monitoring, then tested sensor configurations and signal-processing methods comparing heart-rate, breathing, and movement data for prototype development.
- Conducted customer-discovery research (20 dog owners, 3 veterinarians) to guide product requirements, and designed the data flow between the wearable and mobile app for display and health alerts.

6-DOF Prosthetic Limb, Subteam Leader

Jan 2025 – Dec 2025

TURTLE Robotics, Texas A&M University

College Station, TX

- Enforced documentation and revision control (Anchorpoint) across the CAD team, maintaining a single master file and tracking every design change against the current drawing set to improve coordination.
- Translated project goals into measurable mechanical specifications for the prosthetic-limb prototype, evaluated design alternatives against performance requirements, feasibility, cost, and integration constraints.
- Created and modified SolidWorks parts, assemblies, and drawings to communicate component designs and support design revisions.
- Designed, built, and instrumented actuator-performance test benches following written test procedures, then measured machined actuator and Stewart-platform component dimensions with calipers, tape measures, and gauges to verify parts met engineering-drawing tolerances before leading the sub-team through fabrication and assembly.

Work Experience

Undergraduate Researcher

May 2025 – Jul 2025

Marshall University — NSF REU

Huntington, WV

- Evaluated the reliability of AlphaFold 2 and AlphaFold 3 for membrane-protein structure prediction as

a validation step toward computationally modeling brain-derived neurotrophic factor (BDNF) and its potential interactions with membrane proteins for drug-delivery research.

- Built a Python workflow that compared AlphaFold predictions with experimental structures across approximately 2,400 membrane proteins, creating a consistent dataset for quantitative analysis.
- Used PyMOL and RMSD analysis to compare AlphaFold 2 and AlphaFold 3 outputs; found similar structural agreement, while AlphaFold 2 produced higher alignment identity (88.35% vs. 86.87%) and model confidence (79.341 vs. 77.902).
- Presented weekly research updates to mentors and peers, summarizing experimental results, data-analysis progress, and next steps.

Undergraduate Researcher

Aug 2024 – Dec 2024

Texas A&M University — NES Lab

College Station, TX

- Supported nanoparticle drug-delivery research by reviewing 37 peer-reviewed studies and synthesizing findings into proposed experimental approaches to simulate blood–brain-barrier transport.
- Identified how nanoparticle shell thickness, lattice structure, magnetic-field conditions, ion and protein interactions, and pH may affect computational predictions used to guide experimental design.
- Delivered weekly research updates to the lab’s faculty advisor and graduate mentors, summarizing literature findings and proposed next steps.

Supplemental Instructor Leader, General Chemistry

Aug 2023 – Apr 2024

Texas A&M University

College Station, TX

- Led 3 weekly General Chemistry study sessions for approximately 25 students, explaining course material and guiding collaborative engagement to support student learning.
- Prepared weekly worksheets, review activities, and presentation materials, using feedback from supervisors and peer leaders to adjust session plans and improve student engagement.
- Prepared for weekly sessions by reviewing course content and organizing materials in advance, allowing activities to run on schedule and leaving time to address student questions.

Pharmacy Technician / Pharmacy Customer Associate

Jun 2022 – Nov 2023

Walgreens

College Station, TX

- Processed approximately 300 daily prescriptions in a high-volume retail pharmacy by navigating pharmacy systems, transcribing prescription information, filling medication orders, and preparing completed prescriptions for pharmacist verification.
- Balanced prescription processing, patient communication, and inventory tasks during high-volume periods to maintain an organized pharmacy workflow; trained 3 new hires on workflows and daily procedures.
- Resolved prescription and insurance issues by reviewing patient records, clarifying missing information, and coordinating with pharmacists and patients to keep medication requests moving.